

Mi-4 Cheatsheet:

Recap Tailwind:

A BASICS

Concept	Utility	Example
Font Size	<code>text-xs</code> → <code>text-9xl</code>	<code>text-lg</code>
Font Weight	<code>font-thin</code> → <code>font-black</code>	<code>font-bold</code>
Text Align	<code>text-left</code> , <code>text-center</code> , <code>text-right</code> , <code>text-justify</code>	<code>text-center</code>
Text Color	<code>text-red-500</code> , <code>text-blue-700</code> , etc.	<code>text-green-600</code>
Background Color	<code>bg-red-500</code> , <code>bg-gray-100</code> , etc.	<code>bg-blue-200</code>
Border	<code>border</code> , <code>border-2</code> , <code>border-red-400</code>	<code>border</code> <code>border-gray-300</code>
Border Radius	<code>rounded</code> , <code>rounded-md</code> , <code>rounded-full</code>	<code>rounded-lg</code>
Display	<code>block</code> , <code>inline-block</code> , <code>flex</code> , <code>grid</code> , <code>hidden</code>	<code>flex</code>

B BOX MODEL

Property	Utility	Example
Padding	<code>p-0</code> → <code>p-96</code> , <code>px-4</code> , <code>py-2</code> , <code>pt-3</code>	<code>px-4</code> <code>py-2</code>
Margin	<code>m-0</code> → <code>m-96</code> , <code>mx-2</code> , <code>my-5</code> , <code>mt-3</code>	<code>mt-5</code> <code>ml-4</code>
Width	<code>w-0</code> , <code>w-px</code> , <code>w-1/2</code> , <code>w-full</code> , <code>w-screen</code>	<code>w-1/3</code>

Height	<code>h-0</code> , <code>h-px</code> , <code>h-1/2</code> , <code>h-full</code> , <code>h-screen</code>	<code>h-64</code>
Max Width	<code>max-w-xs</code> → <code>max-w-7xl</code>	<code>max-w-md</code>

C FLEX & GRID

Concept	Utility	Example
Flexbox	<code>flex</code> , <code>inline-flex</code>	<code>flex</code>
Flex Direction	<code>flex-row</code> , <code>flex-col</code> , <code>flex-wrap</code>	<code>flex-col</code>
Justify Content	<code>justify-start</code> , <code>justify-center</code> , <code>justify-between</code>	<code>justify-around</code>
Align Items	<code>items-start</code> , <code>items-center</code> , <code>items-end</code>	<code>items-center</code>
Grid Layout	<code>grid</code> , <code>grid-cols-1</code> → <code>grid-cols-12</code>	<code>grid</code> <code>grid-cols-3</code>
Gap	<code>gap-1</code> → <code>gap-10</code>	<code>gap-4</code>

D POSITION & Z-INDEX

Concept	Utility	Example
Position	<code>static</code> , <code>relative</code> , <code>absolute</code> , <code>fixed</code> , <code>sticky</code>	<code>absolute</code>
Top/Right/Bottom/Left	<code>top-0</code> , <code>left-1/2</code> , etc.	<code>top-4</code> <code>left-4</code>
Z-Index	<code>z-0</code> , <code>z-10</code> , <code>z-50</code> , <code>z-auto</code>	<code>z-30</code>

E EFFECTS

Concept	Utility	Example
Shadow	<code>shadow, shadow-md, shadow-lg, shadow-xl</code>	<code>shadow-md</code>
Opacity	<code>opacity-0 → opacity-100</code>	<code>opacity-75</code>
Mix Blend	<code>mix-blend-normal, mix-blend-multiply</code>	<code>mix-blend-screen</code>
Blur	<code>blur-sm, blur, blur-md, blur-lg</code>	<code>blur-md</code>
Transition	<code>transition, transition-all, duration-300</code>	<code>transition duration-200</code>
Hover	<code>hover:bg-blue-500, hover:text-white</code>	<code>hover:shadow-lg</code>

F RESPONSIVE DESIGN

Breakpoint	Prefix
<code>sm:</code>	<code>≥640px</code>
<code>md:</code>	<code>≥768px</code>
<code>lg:</code>	<code>≥1024px</code>
<code>xl:</code>	<code>≥1280px</code>
<code>2xl:</code>	<code>≥1536px</code>

Example: `md:text-xl, lg:flex, sm:hidden`

G TYPOGRAPHY

Concept	Utility	Example
Font Style	<code>italic, not-italic</code>	<code>italic</code>

Text Transform	<code>uppercase, lowercase, capitalize, normal-case</code>	<code>uppercase</code>
Line Height	<code>leading-none, leading-snug, leading-relaxed</code>	<code>leading-tight</code>
Letter Spacing	<code>tracking-tight, tracking-widest</code>	<code>tracking-wide</code>

H INTERACTIVITY

Concept	Utility	Example
Cursor	<code>cursor-pointer, cursor-wait, cursor-not-allowed</code>	<code>cursor-pointer</code>
Pointer Events	<code>pointer-events-none, pointer-events-auto</code>	<code>pointer-events-none</code>
User Select	<code>select-none, select-text, select-all</code>	<code>select-none</code>

I LIST & TABLE

Concept	Utility	Example
List	<code>list-none, list-disc, list-decimal, list-inside</code>	<code>list-disc</code> <code>list-inside</code>
Table	<code>table, table-auto, table-fixed, table-row, table-cell</code>	<code>table-auto</code>

J BACKGROUND

Concept	Utility	Example
---------	---------	---------

Background Size	<code>bg-cover</code> , <code>bg-contain</code> , <code>bg-auto</code>	<code>bg-cover</code>
Background Position	<code>bg-center</code> , <code>bg-top</code> , <code>bg-bottom</code>	<code>bg-center</code>
Background Repeat	<code>bg-no-repeat</code> , <code>bg-repeat-x</code>	<code>bg-no-repeat</code>

K MISCELLANEOUS

Concept	Utility	Example
Overflow	<code>overflow-auto</code> , <code>overflow-hidden</code> , <code>overflow-scroll</code>	<code>overflow-x-hidden</code>
Object Fit	<code>object-cover</code> , <code>object-contain</code> , <code>object-fill</code>	<code>object-cover</code>
Visibility	<code>visible</code> , <code>invisible</code>	<code>invisible</code>
Aspect Ratio	<code>aspect-auto</code> , <code>aspect-square</code> , <code>aspect-video</code>	<code>aspect-video</code>

L FORMS

Concept	Utility	Example
Input Style	<code>form-input</code> , <code>form-select</code> , <code>form-textarea</code> , <code>form-checkbox</code>	<code>form-input</code>
Focus States	<code>focus:outline-none</code> , <code>focus:ring-2</code>	<code>focus:ring-blue-500</code>



ADVANCED/PLUGIN-BASED

Concept	Plugin Required	Example
Line Clamp	<code>line-clamp-3</code>	Needs <code>@tailwindcss/line-clamp</code>
Aspect Ratio	<code>aspect-[16/9]</code>	Needs <code>@tailwindcss/aspect-ratio</code>
Typography	<code>prose, prose-lg, prose-indigo</code>	Needs <code>@tailwindcss/typography</code>

Tour of DOM (Document Object Model)

♦ What is DOM?

- **DOM (Document Object Model)** is a programming interface for HTML and XML documents.
- It represents the page as a tree structure where each node is an object.
- JS uses DOM to manipulate content, structure, and styles.

♦ `getElementsByTagName(tag)`

- Returns a **live HTMLCollection** of elements with the given tag name.
- Example:

```
const allDivs = document.getElementsByTagName('div');
```

♦ `getElementsByClassName, getElementById`

- **getElementsByClassName('className')**: Returns **live HTMLCollection**.
- **getElementById('id')**: Returns a **single element** with the specific ID.

```
const items = document.getElementsByClassName('item');  
const header = document.getElementById('main-header');
```

♦ **querySelector, querySelectorAll**

- **querySelector(selector)**: Returns the **first matching** element.
- **querySelectorAll(selector)**: Returns a **static NodeList** of all matches.

```
const firstItem = document.querySelector('.item');  
const allItems = document.querySelectorAll('.item');
```

♦ **Dynamic Style, getAttribute, setAttribute, innerText, innerHTML**

Property/Method	Description	Example
<code>element.style.property</code>	Change inline style	<code>div.style.color = "red";</code>
<code>getAttribute()</code>	Get value of an attribute	<code>img.getAttribute('src')</code>
<code>setAttribute()</code>	Set an attribute	<code>img.setAttribute('alt', 'image')</code>
<code>innerText</code>	Gets/sets visible text	<code>para.innerText = 'Hello';</code>
<code>innerHTML</code>	Gets/sets HTML content	<code>div.innerHTML = 'Hi'</code>

♦ **Add/Remove CSS Classes**

```
element.classList.add('highlight');  
element.classList.remove('hidden');
```

```
element.classList.toggle('active');
```

◆ Node vs Collection

Concept	Description
HTMLCollection	Live, returned by <code>getElementsBy...</code>
NodeList	Static, returned by <code>querySelectorAll</code>
parentNode	Returns parent of node
childNodes	Returns NodeList of child nodes (including text)

◆ Create and Append Elements

```
const newDiv = document.createElement('div');  
newDiv.innerText = 'Hello DOM';  
document.body.appendChild(newDiv);
```

Events and Event Handling

◆ What is an Event?

- **Events** are actions users perform (click, input, hover).
 - JS can listen to these events using **event handlers**.
-

◆ Add **onclick** Handler

Inline:

```
<button onclick="handleClick()">Click</button>
```

JS File:

```
document.getElementById('btn').onclick = function() {  
  alert('Clicked');  
};
```

♦ **addEventListener()**

- **Modern and flexible** way to attach event listeners.

```
button.addEventListener('click', () => {  
  alert('Event Listener Clicked');  
});
```

♦ **Common Event Handlers**

Event	Triggered When
click	Click on element
mouseover / mouseout	Mouse enters/leaves
keydown / keyup	Key is pressed/released
submit	Form submission
input	Input field changes
change	Input value changes (after blur)

♦ **Event Bubbling & stopPropagation()**

📌 **What is Event Bubbling?**

- Events **bubble up** from the target element to its ancestors.

```
<div id="parent">
  <button id="child">Click Me</button>
</div>

document.getElementById('parent').addEventListener('click', () => {
  console.log('Parent clicked');
});
document.getElementById('child').addEventListener('click', (e) => {
  console.log('Child clicked');
  // e.stopPropagation(); // stops bubbling
});
```

Output if bubbling is not stopped:

Child clicked
Parent clicked

◆ Event Delegation

✅ What is Event Delegation?

- A technique where a parent handles events of child elements using bubbling.
- Useful for dynamic content.

Example:

```
<ul id="list">
  <li>Item 1</li>
  <li>Item 2</li>
</ul>

document.getElementById('list').addEventListener('click', (e) => {
  if (e.target.tagName === 'LI') {
    alert('Clicked on: ' + e.target.innerText);
  }
});
```

★ Benefits:

- Reduces number of event listeners.
 - Handles dynamically added elements.
-

- Learn multiple ways to handle events.
 - Use `addEventListener` for flexibility.
 - Understand bubbling & delegation to manage events cleanly.
-